IBM DATA SCIENCE CAPSTONE

SPACEX FALCON 9 FIRST-STAGE LANDING ANALYSIS

From launch data to SQL exploration and classification models

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Prepared as a technical project report • Metrics shown are from the notebook results available for this project

Executive summary

Question: can launch data help characterize and predict first-stage landing outcomes?

PROJECT GOAL

Explore Falcon 9 launch history, understand landing outcomes, and compare classification models for first-stage landing prediction.

DATA SNAPSHOT

The modeling dataframe used in the notebook contained 90 rows and 83 features after one-hot encoding.

MODEL SIGNAL

Reported test accuracies: Logistic/SVM/Decision Tree/KNN need to be verified from the final executed notebook; SVM kernel comparison reached up to 83.33%.

- **Workflow:** collect and wrangle data → EDA and SQL → feature preparation → classification → evaluation.
- **Key caution:** the dataset is small, so test accuracy is a useful baseline, not proof of generalization.
- **Deliverables:** machine-learning notebook, SQL notebook, repository documentation, Folium map and Dash dashboard evidence.

Introduction & problem framing

Falcon 9 reusability makes first-stage recovery a meaningful operational question.

BUSINESS / ENGINEERING QUESTION

Can historical launch characteristics help explain landing outcomes and support prediction of a successful first-stage landing?

WHY IT MATTERS

Landing the booster can support reusability and lower the cost of access to space. Outcomes depend on mission, vehicle, payload, orbit, site, and recovery conditions.

- **Scope:** explore launch records and build classification models from structured features.
- **Target:** a landing outcome / success label derived from the capstone dataset; exact encoding should match the executed notebook.
- **Limitations:** historical records, limited sample size, and class imbalance can affect reliability.

Data collection & sources

Source context: IBM Skills Network SpaceX capstone learning labs.

SOURCE

SpaceX launch data prepared for the IBM Data Science Capstone labs; the course provides collection and analysis exercises.

FIELDS

Date/time, booster version, launch site, payload, payload mass, orbit, customer, mission outcome, and landing outcome.

DATA CHECKS

Confirm column names, data types, missing values, duplicate rows, date formats, and target label definitions before analysis.

- Document the original course/source URL and exact dataset version in README.md.
- Upload data files only if redistribution is permitted; otherwise document how to obtain them.
- Keep raw data separate from transformed modeling data to make preprocessing reproducible.

Data wrangling & feature preparation

Preprocessing prepares launch records for EDA and supervised learning.

01

Inspect
Schema, types, missing values

02

Clean
Normalize labels and dates

03

Encode
One-hot encode categorical fields

04

Split
Create training and test sets

05

Evaluate
Use same split for comparisons

MODELING MATRIX

The notebook's one-hot-encoded modeling dataframe was reported as 90 rows × 83 columns.

REPRODUCIBILITY NOTE

The notebook scales features before the split in the current workflow; for a stricter evaluation, fit preprocessing on training data only to avoid leakage.

EDA & interactive visual analytics methodology

Use static summaries to identify patterns, then interactive tools to explore geography and launch records.

DISTRIBUTIONS

Counts by launch site, orbit, booster version, customer, mission outcome and landing outcome.

RELATIONSHIPS

Compare landing outcomes with launch site, payload mass, orbit and time to identify potentially informative features.

INTERACTIVE VIEWS

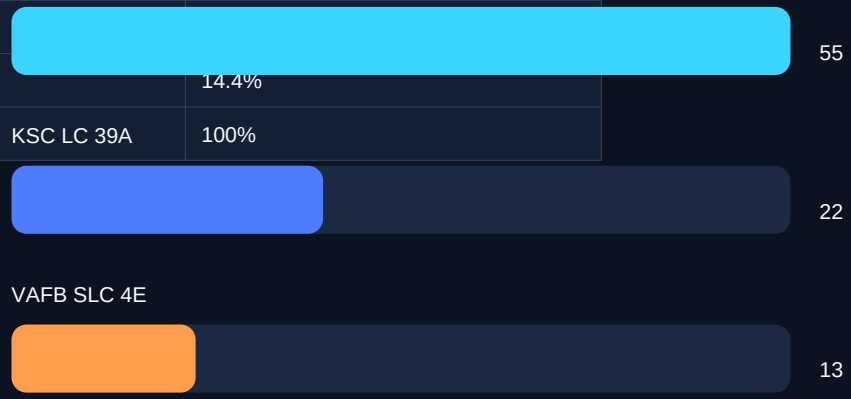
Folium map for launch-site geography; Plotly Dash for filterable launch and outcome exploration.

- Use consistent category labels and show counts as well as percentages.
- Separate descriptive association from causal claims: EDA shows patterns, not proof of causes.
- Capture actual outputs from the running map and dashboard for the submitted evidence.

EDA result: launch-site distribution

Launch-site counts reported from the project analysis.

Launch site	Launch records	Share of 90
CCAFS SLC 40	55	61.1%
KSC LC 39A	22	24.4%
VAFB SLC 4E	13	14.4%
Total	90	100%



Insight: the dataset is concentrated at CCAFS SLC 40. Results may reflect site distribution as well as landing dynamics; inspect per-site outcome rates rather than raw counts alone.

EDA result: orbit & landing outcome mix

Counts available from the project analysis; outcomes are not all binary.

Orbit	Count
GTO	27
ISS	21
VLEO	14
PO	9
LEO	7
SSO	5
MEO	3
Other listed orbits	4

Landing outcome	Share
True ASDS	45.6%
None None	21.1%
True RTLS	15.6%
False ASDS	6.7%
True Ocean	5.6%
Other listed outcomes	5.5%

Interpretation: ASDS denotes autonomous spaceport drone ship. The label distribution is imbalanced and includes no-attempt / unspecified outcomes, so the target definition must be stated clearly before model evaluation.

SQL exploratory analysis methodology

Queries executed from Python with pandas and the existing SQLite connection `con`.

- **Distinct sites:** SELECT DISTINCT "Launch_Site" FROM SPACEXTBL;
- **Prefix filter:** WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
- **Payload total:** SUM("PAYLOAD_MASS__KG_") with customer filter for NASA (CRS).
- **Earliest ground-pad success:** MIN("Date") filtered by 'Success (ground pad)'.
- **Maximum payload booster:** subquery using MAX("PAYLOAD_MASS__KG_").
- **2015 drone-ship failures:** substr("Date",6,2) for month and substr("Date",0,5)='2015' for year.

SQL findings & reproducible evidence

SQL tasks establish data quality and summarize patterns before predictive modeling.

FILTERING

LIKE and date substrings isolate records by launch-site prefix and calendar year.

AGGREGATION

SUM, COUNT, MIN, MAX, GROUP BY and ORDER BY summarize payloads and outcomes.

SUBQUERY

Compare each payload mass to the maximum payload mass to return all matching booster versions.

Evidence to insert before submission: a real screenshot of the SQL notebook showing query code and output tables. Numeric results are intentionally not fabricated here; run the notebook and retain the actual outputs.

Interactive map results — Folium

Map should connect launch-site locations with the available launch records.

MAP CONTENT

Pins for CCAFS SLC 40, KSC LC 39A and VAFB SLC 4E using verified coordinates.

INTERACTION

Popups can show launch-site name and counts; optional layers can separate sites.

DELIVERABLES

Upload Python source and exported HTML map; include a screenshot of the actual map opened in a browser.

SCREENSHOT PLACEHOLDER — replace with your actual Folium HTML map view before final submission.

Do not submit this placeholder as proof of a working map.

Interactive dashboard results — Plotly Dash

The dashboard should allow evaluators to explore the data, not just view a static chart.

FILTERS

Launch site, orbit, booster / outcome filters where supported by the dataset.

VISUALS

Launch counts and outcome summaries; include axis labels, legends, and informative hover text.

RUN INSTRUCTIONS

Document environment setup, run command, local URL, required data files, and dependencies in README.

SCREENSHOT PLACEHOLDER — replace with your actual running Dash app, including visible filters and charts.

Do not submit this placeholder as proof of a working dashboard.

Predictive analysis methodology

Classification models compare alternative decision boundaries on the prepared launch features.

TARGET & FEATURES

Use the landing-success target and documented input features from the executed notebook. Confirm the positive class and label mapping.

MODELS

Logistic Regression, Support Vector Machine (SVM), Decision Tree and K-Nearest Neighbors (KNN).

TUNING

GridSearchCV with cross-validation searches hyperparameters; compare on the same held-out test split.

- Report test accuracy together with confusion matrix and per-class precision/recall where possible.
- A small, imbalanced dataset can make accuracy misleading; inspect class-level metrics.
- Avoid data leakage: fit scaling/encoding on training data only and then transform the test set.

Predictive classification results

Reported notebook metrics available so far; final values should be reconfirmed after a complete run.

Experiment	Test accuracy	Interpretation
SVM — linear kernel	66.67%	Lowest of the three tested kernels
SVM — RBF kernel	77.78%	Improved over linear
SVM — sigmoid kernel	83.33%	Highest among reported SVM kernel tests
Decision Tree — default	88.89%	Reported default run; not confirmed as tuned best model

KEY TAKEAWAY

Kernel choice changes test performance; compare models with consistent splits and metrics.

REPORTING CAUTION

Do not claim a final model winner until all tuned model scores have been run and checked.

Conclusion & innovative insights

Translate descriptive findings into useful next steps without overstating what a small dataset can prove.

WHAT WE LEARNED

The dataset is site- and outcome-imbalanced. SQL provides concise summaries; ML results vary by model and SVM kernel.

INNOVATIVE INSIGHT 1

Compare landing-success rates by site and orbit, not only launch volume; raw counts can be dominated by the most-used site.

INNOVATIVE INSIGHT 2

Use calibrated probabilities and per-class recall to understand uncertainty, particularly for rare landing outcomes.

- Extend the work with time-aware validation, richer mission/vehicle features, and more launch records.
- Build a reproducible pipeline that stores preprocessing and model parameters together.
- Present Folium and Dash outputs as interactive evidence, with clear filters and readable charts.

Final deliverables & repository

Keep the repository public and make the evaluator's review path obvious.

GITHUB REPOSITORY

https://github.com/satejgujar198/SpaceX-Capstone

Upload notebooks, map source + HTML, Dash app, requirements, evidence screenshots, README and final PDF.

FINAL PRE-SUBMISSION CHECK

Run notebooks • verify outputs • add real map/Dash screenshots • confirm data permissions • check repository access • upload this PDF.